Power equations

1

a $x = 3$ **b** $x = 8$ **c** $x = \frac{8}{5}$ **d** $x = 6$

e $x = 2$ **f** $x = 6$ **g** $x = 5$ **h** $x = 2$

Exponential equations

2

a $x = 8$ **b** $x = \frac{2}{9}$ **c** $x = 3$ **d** $x = 0$

e $x = -2$ **f** $y = -3$ **g** $x = -3$ **h** $x = \frac{1}{3}$

i $n = \frac{1}{3}$ **j** $x = -2$

3

a

i $2^{3x} = 2^2$ **ii** $x = \frac{2}{3}$

b

i $2^{4x} = 2^{-1}$ **ii** $x = -\frac{1}{4}$

c

i $2^{-10} = 2^{2x}$ **ii** $x = -5$

d

i $2^{\frac{x}{2}} = 2$ **ii** $x = 2$

4

a $y = 4$ **b** $k = -2$ **c** $y = \frac{3}{2}$ **d** $x = 2$

e $x = 2$ **f** $x = \frac{7}{3}$ **g** $x = -7$ **h** $x = \frac{21}{19}$

i $y = \frac{11}{6}$ **j** $x = -\frac{41}{6}$ **k** $x = 5$ **l** $x = 1$

m $x = 1 + \frac{2k}{3}$ **n** $x = 5$ **o** $x = \frac{5}{2}$ **p** $x = 4, x = -1$

q $x = 3$ **r** $x = \frac{4}{1+n}$

5

a

i $m = 2^x$ **ii** $x = 3,\ 0$

b

i $m = 2^x$ **ii** $x = 3,\ 2$

c

i $m = 2^x$ **ii** $x = -1,\ 3$

d

i $m = 4^x$ **ii** $x = 0,\ 3$

e

i $m = 2^x$ **ii** $x = 0,\ 2$

f

i $m = 3^x$ **ii** $x = 1,\ 2$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Exponential equations and logarithms

6

a $3 < x < 4$

b $3 < x < 4$

c $-4 < x < -3$

d There is no real solution.

7

a

i $x = \dfrac{\log 5}{\log 13}$ ii 0.627

b

i $x = \dfrac{\log \frac{1}{11}}{\log 5}$ ii -1.490

c

i $x = \dfrac{\log 2}{\log 3}$ ii 0.631

d

i $x = \dfrac{\log 6.4}{\log 4}$ ii 1.339

e

i $x = \dfrac{\log 5}{\log 0.4}$ ii -1.756

f

i $x = \dfrac{\log 3125}{\log 5}$ ii 5

g

i $x = \dfrac{\log 6}{\log \frac{1}{2}}$ ii -2.585

h

i $x = \dfrac{\log 19\,207}{\log 27}$ ii 2.993

8

a $x = \dfrac{\frac{\log 70}{\log 4} + 8}{2}$

b 5.532

9

a

i Step 2, when bringing the a out of the index it should multiply the logarithm and not add to it.

ii $a = \dfrac{\log 40}{\log 9}$

iii 1.679

b

i Step 3. The 2 and 89 are in the incorrect positions in the logarithm.

ii $a = \dfrac{\log 89}{\log 2}$

iii 6.476

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Applications

10

a $t = 5$ hours **b** $t = 10$ hours **c** $t = 11.55$ hours

11

a 500 mice **b** 20% **c** 6.03 months

12

a 800 wallabies **b** 15% **c** 8.53 years

13 6.180 hours

© ClipSAT: By Mr. Mohamed Abdallah +966597688647